George Zhang

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Canadian & Hong Kong citizen · eligible for U.S. internships (J-1); TN-eligible post-graduation

Education

University of Waterloo — Waterloo, ON

Bachelor of Computer Science (Honours, Co-op) — Artificial Intelligence Specialization — *Sep 2024 – May 2029*

- Relevant coursework: Optimization, Probability & Statistics, Combinatorics, Data Structures & Algorithms, OOP (C++), Computer Organization.

Experience

Ford Motor Company — Waterloo, ON

Software Developer Intern — IVI Phone Apps — *May 2026 – Aug 2026*

- Shipped features across three in-vehicle apps (Dialer, Settings, Messenger) on Android Automotive OS (AAOS), Ford's production infotainment platform, in a multi-module Java/Kotlin codebase; closed all 64 completed issues, a 100% completion rate on finalized work.
- Spearheaded two automated workflows for the team's LLM log-analysis playbook: tag-differential-and-chunking updates and a direct Confluence sync, removing manual upkeep from the process.

University of British Columbia — Vancouver, BC

Undergraduate Research Assistant — Centre for Brain Health — *Jun 2025 – Aug 2025*

- Led development of two full-stack hand-rehabilitation apps within PIKA, an AI Parkinson's-care platform deployed at Vancouver Coastal Health.
- Built a real-time computer-vision pipeline (Python, OpenCV, MediaPipe) detecting 7 gestures at ∼90% accuracy with under 100 ms latency, plus dashboards that cut clinician monitoring overhead by 30%; presented at the 11th Singapore International Parkinson Disease & Movement Disorders Symposium.

Projects

Limit Order Book & Matching Engine | C++20 — github.com/TheYellowDuck/limit-order-book

- Architected a low-latency, lock-free matching engine (strict price-time priority, 9 order types + pro-rata) on custom cache-optimized structures — a two-level hierarchical-bitmap price ladder and intrusive order pool give O(1) level access and O(1) cancels via hardware bit-scan, with integer-tick prices for exact, deterministic matching.
- Engineered a wait-free runtime (SPSC ring buffer, single-writer matcher, seqlock market-data publishing) with deterministic journaling/replay; validated with differential, property-based, and fuzz testing under sanitizers.

Code-Aware RAG System | Python, tree-sitter, embeddings — github.com/TheYellowDuck/RAG-codebase

- Engineered a retrieval system fusing dense + BM25 (Reciprocal Rank Fusion) with a code graph and personalized PageRank, answering codebase questions with citation-backed answers.
- Treated each design choice as a measured ablation — the LLM reranker significantly improved ranking (MRR/NDCG, $p < 0.001$) while a cross-encoder was net-negative on code — with recall@k, bootstrap CIs, and paired significance tests.

Autonomous LLM Web Agent | Python, Playwright — github.com/TheYellowDuck/web-agent

- Designed a ReAct + reflection agent over a browser accessibility tree with a WebArena / Online-Mind2Web eval harness (Wilson CIs, pass@k, failure taxonomy).
- Isolated measurement from capability: the same agent scores 0.43 vs 0.61 on WebArena by scorer alone (substring vs type-aware) — a 100%-one-directional disagreement (kappa = 0.66).

Technical Skills

Languages	C++, Python, Java, C, Kotlin, TypeScript/JavaScript, SQL
Systems	Lock-free / wait-free concurrency, cache-aware data structures, multithreading, low-latency design
Math & ML	Statistics, significance testing, probability, NumPy, Pandas, PyTorch
Tools	Git, Linux, CMake, Docker, GitHub Actions, sanitizers, gdb
Concepts	Data Structures & Algorithms, Benchmarking & Evaluation, Systems Programming, OOP

Awards & Competitive Programming

- President's Scholarship of Distinction & Entrance Scholarship in Mathematics, University of Waterloo
- ICPC Local (Waterloo) — Rank 28 / ∼100 · 470+ LeetCode & 220+ DMOJ solved (1,497 points)
- Canadian Computing Competition — Junior: perfect score, Senior: top quartile · AIME Qualifier · COMC Distinction
- RoboCup International Junior Maze Rescue 2023 (Captain) — 2nd Place Supergroup, 17th Individual